EUAEL MEKONEN ESHETE

euaelmeko@gmail.com · github.com/euaell · linkedin.com/in/euael-eshete · portfolio.euaell.me · Addis Ababa, Ethiopia

Software engineer working on PKI and multi-tenant SaaS at Securetron. I build the licensing, tenant isolation, certificate lifecycle, and the auth around them. Most of my work is C# and .NET, with Python, Go, and TypeScript where they fit. I also run my own infrastructure and a small software studio, ZafTech.

WORK EXPERIENCE

Software Developer · Securetron *Mar 2025 – Present · Addis Ababa / remote*

PKI Trust Manager is a multi-tenant certificate lifecycle management (CLM) SaaS.

- Built the licensing engine that handles online and offline activation, grace periods, certificate limits, and read-only mode when a license lapses. Offline licenses verify with a signature check, so the product needs no server call.
- Led the multi-tenant refactor. Licenses load per organisation, queries scope to the customer, and middleware enforces tenant scope.
- Shipped DigiCert CertCentral end-to-end: orders, webhooks, API keys encrypted at rest, and a feature-flagged rollout. Integrated EJBCA and hardened its credential handling. Wrote CA abstractions so a new provider plugs into the existing flows.
- Built an assistant inside the product. It stores conversations, respects roles, writes to the audit log, and uses versioned system prompts.
- Moved the portal from legacy CSS to Tailwind, then to server-rendered HTMX and Alpine.js partials.
- Set up Azure Pipelines and Docker builds for every service, and Azure Bicep for multi-tenant provisioning.

Stack: *ASP.NET Core · EF Core · SQL Server · HTMX · Alpine.js · Tailwind · Azure Pipelines · Bicep · Docker*

Founder · ZafTech *2025 – Present · zaftech.co*

A small software studio. I design, build, and host client work and our own products on a self-managed stack.

- Products include Convia (conversational forms with strict schema compilation) and Mizan (nutrition and training tracking with first-class Model Context Protocol integration).

Stack: *Astro, Next.js, Tailwind, Bun, Node.js, Go, PostgreSQL and PostGIS, Redis, BullMQ, Docker, BetterAuth*

Software Developer · AfroChat (A2SV) *Sep 2023 – Mar 2025 · Addis Ababa*

- Built async backend APIs in FastAPI and Litestar, and shipped features to real users.
- Moved media assets to Google Cloud Storage with no downtime and rewrote asset delivery, dropping infrastructure costs by 94%.
- Built a retrieval-augmented generation (RAG) pipeline with a vector database and the OpenAI API.
- Moved the monolith to containers on Cloud Run, with autoscaling on CPU and request latency.

Stack: *Python · FastAPI · Litestar · GCP · Cloud Run · PostgreSQL · Docker*

Full-Stack Intern · New Era Research and Development *Mar 2022 – Jun 2022 · Addis Ababa*

- Built a control prototype for a 3-axis CNC machine on Arduino.
- Built an internet onboarding app with C# and ASP.NET.

PROJECTS

DistroKV (In Progress) *Go · Raft · Kubernetes*

A distributed key-value store. Raft handles leader election and replication. An LSM-tree engine handles storage. The cluster runs across three regions to test real network partitions.

EDUCATION

BSc in Computer Engineering · Addis Ababa University *Sept 2018 – Jun 2023 · Addis Ababa, Ethiopia*

- **Major GPA: 3.81/4.00** · Graduated with High Distinction
- **Thesis:** IoT-based Water Distribution Monitoring System – Developed an IoT solution to monitor and optimise water distribution in resource-constrained environments.
- **Relevant Coursework:** Database Systems, Digital Logic Design, Digital Signal Processing, Software Engineering, Computer Architecture and Organisation, Object-Oriented Programming

TECHNICAL SKILLS

- **Languages:** C#, Python, Go, TypeScript, SQL
- **Backend:** ASP.NET Core, EF Core, FastAPI, Litestar, Node.js, gRPC, REST
- **Frontend:** HTMX, Alpine.js, React, Next.js, Tailwind
- **Security:** PKI and X.509, CA integration (DigiCert, EJBCA), mTLS, OAuth 2.0 & OIDC, secrets handling
- **Infrastructure:** Docker, Azure Pipelines, Azure Bicep, GCP (Cloud Run, Cloud Storage), GitHub Actions, Linux, Nginx, Kubernetes (learning)
- **Data & AI:** PostgreSQL, SQL Server, Redis, vector databases, RAG, embeddings, LLM tool use, prompt versioning

- **Spoken Languages:** Amharic (Native), English (Fluent / IELTS 7.5), German (Beginner)

MENTORING & LEADERSHIP

Mentor · Africa to Silicon Valley (A2SV) *Sep 2023 – Mar 2025*

Mentored more than 150 students in full-stack development, LLM integration, and cloud.

Mentor · Solvers Club, Addis Ababa University *Mar 2022 – Jun 2022*

Helped first-year students with university life and STEM courses.